

Math/Stat 4520: Mathematical Statistics

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Engineering Outreach

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Class Room: Wherever you might be...

Course Description

In this course, we will focus on understanding the interplay between probability, statistics, and modeling. In many regards, probability and statistics are complementary branches of applied mathematics. Probability expresses the chance of observing an outcome given a model (e.g., a distribution); conversely, statistics asks whether a given set of observations are consistent with a parameterization of a chosen mathematical model given an assumed probability distribution. More specifically, in statistics we often seek to find a range of parameter values in which we are confident or to distinguish between competing models with some degree of certainty. Likewise, “inverse problems” in mathematics attempt to find valid parameters given a generative model and observed data, which is simply another way to discuss the practice of statistical model fitting. This course will help you understand and implement the tools required to build complex models and fit them to real-world data.

Course Structure

This class is an online class, and, as such, content delivery will be done via pre-recorded videos. As such, progress in the course is up to you, however, homework and exams *have specific due dates*. **You must turn in the homework and exams on-time in order to get credit for them. Late work will not be accepted.** See the section below regarding engineering outreach due dates.

Coursework will be as follows. Suggested practice problems will be given for each section of the book. Practice problem sets will be posted to Canvas. Answers to the suggested problems can be found in the back of the book. Each Friday there will be *assigned* homework (these are “quizzes” in the in-person version of the class). These homeworks will only be spot checked and marked for completion. (In other words, I will not carefully grade each question.) In general, I expect you to get 100% of the points available from this portion of the class. You may use whatever resources you can find to help you answer the questions (the book, lectures, AI, etc.). However, I **strongly** recommend that when attempting these assignments you (1) do so without the help of any resources and (2) give yourself 15–20 minutes to answer the problems. After you have attempted this, you can go back to do corrections/further study your answers. Why is this

important? First, it familiarizes you with the style of questions that I write. Second, it forces you to work under time pressure and see how long it takes to answer my questions. These two items are *extremely* important when it comes to taking the exams; if you do the assigned homeworks (“quizzes”) in this manner it will help you prepare for a timed exam written by me. Furthermore, it actually forces you to attempt to answer problems without any aid (as will be the case when taking the exams).

The graded portion of the class will be your performance on three equally-weighted, non-cumulative exams, in addition to the assigned homework. I expect to give 9 assigned homework sets (again, these are called quizzes in the normal class and will be titled as such). You can drop one of the assigned homework sets (so 8 quizzes count toward your grade). The assigned homework will cumulatively count as one exam. In other words, each evaluated facet of class is worth 25% of your overall grade. Chapters of the textbook are fairly short. Exams will roughly cover about 9 chapters. *You will be allowed to have a page of notes during each exam, but no calculators are allowed!* **If you need an extension in the due date for an assigned homework or exam, arrangements need to be made with me prior to the event.**

Required Materials

- *Probability and Mathematical Statistics*. By MC Meyer. Society for Industrial and Applied Mathematics: Philadelphia.

Prerequisites/Corequisites

Prerequisites: Math/Stat 4510

Topics By Week (*Approximate*)

Note that session numbers (e.g., session 1 is noted §1) are given for EO students and are also approximate. *As mentioned above, quizzes are given at the beginning of class of Fridays.*

1. Wks 1–2; §1–4: Introduction and review of distributions (discrete, continuous, multivariate)
2. Wk3; §5: R, §6: Ch. 42 (Parameter Estimation), §7: Ch. 43 (MLE)
3. Wk 4; §8: Ch. 43 (MLE cont.), §9 & §10: Ch. 44 (MM; W&F)
4. Wk 5; §11: Review, **Test 1**, §12: Ch. 46 (Consistency)
5. Wk 6; §13: Ch. 50 (CIs), §14: Ch. 51 (Bootstrapping), §15: Ch. 52 (Information)
6. Wk 7; §16: Ch. 53 (Sufficient Statistics), §17: Ch. 54 (UMVUE)
7. Wk 8; §18: Ch. 55 (Exponential Family), §19: Numerical methods, §20: Review
8. Wk 9; **Test 2**, §21 Test 2 Summary
9. Wk 10; *Spring Break*

10. Wk 11; §22: Ch. 34 (Sampling a Normal), §23: Ch. 35 (Hypothesis Testing), §24: Ch. 36 (Hypothesis Testing)
11. Wk 12; §25: Ch. 56 (Size and Power), §26: Ch. 57 (Neyman-Pearson Lemma), §27: Ch. 58 (LRT)
12. Wk 13; §28: Ch. 58 (LRT cont.), §29: CIs and testing, §30: Review
13. Wk 14; **Test 3**, §31 & §32: Ch. 59 (χ^2 -tests; W&F)
14. Wk 15; §33 & §34: Models relying on normal distributions (M&W), §35: Ch. 60 (ANOVA)
15. Wk 16; §36: Ch. 45 (Bayesian Estimation), §37 & §38: Ch. 49 (Credible Intervals; W&F)
16. Wk 17; §39: Model selection, Hierarchical models, §40: MCMC, §41: Review

Course Objectives/Learning Outcomes

Successful students will:

1. Understand the theory of statistics sufficiently to solve most problems appearing in standard textbooks for this course.

Final Exam Date

The fourth exam will *be due* by the last Friday of final exam exam (May 15th). **I will not accept the fourth exam past this date.**

Grading Policy

A typical grading scale will be used with cutoffs every 10% (i.e., $\geq 90\% = A$, $(90\%, 80\%] = B$, ...). I reserve the right to curve the scale dependent on overall class scores at the end of the semester. Any curve will only ever make it easier to obtain a certain letter grade. The grade will count the assessments using the following proportions:

- **25%** of your grade will be determined by 3 of the 4 exams (lowest exam by percentage is dropped).
- **25%** of your grade will be determined by 8 of your 9 assigned homework scores (lowest score is automatically dropped).

Engineering Outreach Due Dates

All assigned homework ("quizzes") and exams will be due one week after their actual delivery in class. For example, if a test was given on a Wednesday, EO students will have until the next Wednesday to have a proctored exam taken and returned for grading.